

Response to Reviewers

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Title: Explainable AI for Student Success: A Multi-Objective Framework with SHAP-Based Personalized Recommendations.

Authors: Shikha Pachouly, D. S. Bormane.

Letter to the Editor and Reviewers

We sincerely thank the Editor and both Reviewers for their thorough, constructive, and insightful feedback. The comments significantly improved the methodological rigor, clarity, transparency, and overall positioning of the manuscript. In response to the reviewers' suggestions, we performed extensive revisions throughout the paper.

Beyond point-by-point textual corrections, the revised manuscript incorporates several substantial methodological and experimental enhancements. Specifically, we: (i) implemented six additional evaluation and validation components, including a leakage-free ablation study, subgroup fairness audit, causal plausibility analysis via propensity-score matching, and reimplementation-based benchmarking against three recent 2024–2025 state-of-the-art studies; (ii) regenerated all reported numerical results under a unified canonical 5-fold stratified cross-validation protocol to ensure consistency and reproducibility; and (iii) added five new manuscript sections (Sections 3.10, 4.3, 5.4, 5.5, and 5.7) to improve methodological transparency and experimental completeness.

All modifications made in the revised manuscript are highlighted in red font in accordance with the Editor's instructions. In the responses below, we reproduce each reviewer comment, describe the corresponding revisions and actions taken, identify the relevant manuscript locations (section/page references), and report updated quantitative results where applicable.

The reproducible implementation and experimental code are available at:

<https://github.com/shikhapachouly/student-recomondation>

We hope the reviewers and editor will find the revisions responsive, transparent, and complete. Sincere thanks once more for the time and care invested in reviewing our work.

Shikha Pachouly and D. S. Bormane

1 Response to Reviewer 1

[R1.1] Drawbacks of conventional techniques and positioning of this work

“The problem definition of this work is not clear. In Sect. 2, the drawbacks (Not limitation = research depth) of each conventional technique should be described. The authors should emphasize the difference with other methods to clarify the position of this work further.”

Action. Section 2 has been substantially expanded. For each cited technique we now state (i) what the method does, (ii) its specific drawback in the context of multi-objective student-success modelling, and (iii) how our work resolves that drawback. A new closing paragraph at the end of Section 2.1, Section 2.2 and Section 2.3 explicitly positions our work against the five identified gaps.

Evidence Three new paragraphs have been inserted at the ends of Section 2.1, Section 2.2, and Section 2.3 respectively:

- In Section 2.1, gradient-boosted/ensemble pipelines are critiqued for treating student outcomes as a single scalar, conflating academic performance with at-risk status and career readiness, and offering no automatic translation from feature attribution to advisor-facing recommendation.
- In Section 2.2, existing XAI-in-EDM systems are shown to stop at *post-hoc explanation*: SHAP plots are produced but no system automatically converts attributions into actionable, personalised recommendations or audits them for fairness across demographic subgroups.
- In Section 2.3, modern tabular-DL architectures (TabNet, FT-Transformer, SAINT) are critiqued for relying on large feature interactions that small educational survey datasets (< 2,000 records) cannot reliably support, leading practitioners to over-invest in DL architectures rather than feature engineering.

[R1.2] Notation list

“To help readers’ understanding, the authors should add a notation list, because there are many variables in equations.”

Action. A complete **Notation Table 2** has been added at the end of Section 3.9. Every symbol used across Eqs. (1)–(19) is enumerated with its meaning.

Evidence The new table contains below entries:

$N, d, x_i, g_i, b_i, t_i, p_i, y_i^{(k)}, C_k, w_c, \phi_j, \lambda, \gamma, F_{\text{mod}}, F_{\text{ctx}}, F_{\text{imm}}, \sigma_i, h_i, r_j, T, K, \text{TPE}, \text{ATT}, \text{FNR}.$

[R1.3] Figure quality (Figs 3, 5, 7, etc.)

“In figures, letters are small and blurry. The authors should enlarge or redraw figures.”

Action. Wide figures like (Figs. 3, 5–7, 9 etc) are now in one-column page-width layout as per review comment.

[R1.4] Comparison with state-of-the-art (2024–2025)

“The reviewer fails to understand the effectiveness of the proposed technique, because no comparison data with state-of-the-art methods is shown in this paper. ... In-depth comparison and discussion with state-of-the-art methods proposed in 2024-2025 are necessary to clarify the effectiveness.”

Action. A new **Section 5.4 “Comparison with Recent State-of-the-Art XAI–EDM Methods”** has been added, accompanied by **Table 9**. Three representative 2024–2025 SHAP-based studies were re-implemented on *our* dataset under our *identical* 5-fold stratified CV protocol (SMOTE on training folds only, seed 42). The implementations are released as part of the code in the github repository (`revision/code/sota_baselines.py`).

Studies evaluated.

1. **Abukader et al. 2025 [16]** — metaheuristic-tuned LightGBM with class-balanced loss and SHAP explanation.
2. **Liu et al. 2025 [15]** — stacking ensemble (Random Forest + Gradient Boosting + XGBoost base learners; logistic-regression meta-learner) with SHAP.
3. **Kalita et al. 2025 [13]** — bidirectional LSTM treating the feature vector as a length- d sequence (the adaptation Kalita et al. apply to cross-sectional inputs), with SHAP via KernelExplainer.

Quantitative result (Table 9).

Method	CGPA	At-Risk	Career
Abukader 2025 [16] – LightGBM + SHAP	0.455 ± 0.014	0.637 ± 0.029	0.692 ± 0.0
Liu 2025 [15] – Stacking ensemble + SHAP	0.448 ± 0.026	0.612 ± 0.028	0.640 ± 0.0
Kalita 2025 [13] – Bi-LSTM + SHAP	0.308 ± 0.038	0.559 ± 0.042	0.518 ± 0.0
This work – XGBoost + engineered features + SHAP recs	0.805 ± 0.017	0.767 ± 0.040	0.699 ± 0.0

[R1.5] Feature-count inconsistency (50+/43/40)

“The manuscript states that the dataset contains “50+ features” in the Abstract, but later states that “43 features span seven categories.” Table 2 also reports category counts summing to 40, not 43.”

Action. All three inconsistencies have been **corrected** in the revised manuscript, and the numerical reconciliation has done. The Abstract, Section 3.2, Section 3.9, Section 4.1 and Table 4. The recommendation taxonomy have all been re-stated using the audited values, and they now agree on a single, consistent count of **42 model-input features**.

Evidence (full reconciliation). Out of the **52 raw** survey columns:

$$52 - \underbrace{4}_{\text{IDs}} - \underbrace{7}_{\text{target-derived}} - \underbrace{4}_{\text{low-variance}} + \underbrace{5}_{\text{engineered}} = 42 \text{ model-input features.}$$

[R1.6] Table 2 row-count inconsistency ($3+3+9+10+8+3+4 = 40 \neq 43$)

“Table 2 lists feature-category counts as $3 + 3 + 9 + 10 + 8 + 3 + 4 = 40$, while the Total row reports 43. This is a direct numerical inconsistency.”

Action. Table 4 (after updates and additon of new tables) has been replaced with a corrected version . The engineered features now appear as a separate row, and the totals reconcile exactly.

Evidence (corrected Table 2).

Category	Count	Example features
Academic	4	tenth_percentage, twelfth_percentage, course_branch, nursery_education
Demographic	3	gender, residence_type, institute_name
Family/Socio-economic	9	mother_education_level, mother_occupation, family_income_bracket
Behavioural	9	daily_study_hours, workshop_seminar_participation_yn, community_service_yn
Skills	5	communication_skills_scale_1to5, leadership_scale_1to5
Wellbeing	3	health_status_scale_1to5, stress_frequency_scale_1to5, stress_coping_yn
Choice context	4	college_choice_reason, course_choice_reason, internet_access
Engineered	5	academic_index, skills_index, engagement_score, study_stress_interaction, skills_study_interaction
Total	42	

Verification: $4 + 3 + 9 + 9 + 5 + 3 + 4 + 5 = 42$.

[R1.7] Recommendation taxonomy: $16 + 6 + 19 = 41 \neq 43$

“The recommendation-feature taxonomy reports 16 modifiable, 6 contextual, and 19 immutable features, which sum to 41, not 43.”

Action. The taxonomy in Section 3.9 and Section 4 has been recounted and reconciled against the feature matrix, please refer Table 4.

Evidence.

Tier	Count
Modifiable	12
Contextual	3
Immutable	22
Engineered	5
Total	42

Verification: $12 + 3 + 22 + 5 = 42$.

[R1.8] “3.800d7” artefact in Section 5.5

“The text states that `academic_index` with $SHAP = 0.126$ is “3.800d7” more influential than `tenth_percentage` with $SHAP = 0.033$. The numerical ratio is approximately 3.82.”

Action. Confirmed as a Word/Equation-Editor export artefact. Replaced with the correct ratio.

Evidence The sentence now reads: “*academic_index* ($SHAP = 0.126$ for at-risk) is **approximately $3.82 \times$ more influential** than the next raw feature (*tenth_percentage*, $SHAP = 0.033$).” Please refer section 5.8.

[R1.9] “6.600d7” artefact in Section 5.5 Discussion

“Workshop participation with $SHAP = 0.331$ has a $SHAP$ impact “6.600d7” larger than teamwork skills with $SHAP = 0.050$. The numerical ratio is approximately 6.62.”

Action. Same Equation-Editor export artefact. Corrected.

Evidence (manuscript). “The highest-impact intervention (workshop participation) has a $SHAP$ impact **approximately $6.62 \times$ larger** than the fifth-ranked factor (teamwork skills).” Please refer section 5.8.

[R1.10] Transformer F1 range (0.41–0.58 vs. Table 4 0.423–0.584)

“The manuscript states that transformer-based models have $F1 = 0.41\text{--}0.58$, but Table 4 shows the relevant FT-Transformer and SAINT $F1$ -macro values range from 0.423 to 0.584.”

Action. The prose range in Section 5.2 has been rounded consistently with Table 7 (Table updated from 4 to 7 due to new tables insertion) .

Evidence (manuscript). “Transformer-based models struggle: FT-Transformer and SAINT significantly underperform on all objectives ($F1 = \mathbf{0.42\text{--}0.58}$).” Please refer section 5.2.

2 Response to Reviewer 2

[R2.1] No reproducible comparison with 2024+ XAI-based methods on a common protocol

“The literature table lists recent methods, but the experimental section evaluates only generic baselines and tabular architectures. ... The authors should either remove superiority implications or add reproducible comparisons with recent SHAP-based ensemble, Light-GBM/XGBoost, imbalance-learning, and recommendation-oriented EDM methods under the same evaluation protocol.”

Action. Addressed jointly with [R1.4] in the new Section 5.4. Three State Of The Art studies are reimplemented and benchmarked under our protocol; their source code is part of the public replication package (`revision/code/sota_baselines.py`). Superiority implications have been softened throughout: the contribution of this work is now framed as the *unified multi-objective + automated recommendation* pipeline, not as a categorical claim of architectural superiority.

[R2.2] SHAP is associational, not causal

“The recommendation engine treats SHAP attribution as if it can support intervention prioritization, but SHAP only explains model predictions and does not establish causal effects. ... The authors should reframe recommendations as hypothesis-generating decision support and add causal validation, propensity-score analysis, ablation, or prospective intervention evidence.”

Action. This is an important and well-taken point. We have responded with substantive changes:

1. **Reframing across the manuscript.** The Abstract, Section 3.9 caption, Section 5.8 have all been revised to describe recommendations as *hypothesis-generating decision support for academic advisors*, never as validated causal interventions. Phrasing such as “high-leverage intervention” has been replaced by “candidate intervention.”
2. **New Section 5.9 Causal Plausibility via Propensity-Score Matching.** For the three highest-ranked modifiable features (workshop participation, high peer-group quality, stress coping), we estimate the Average Treatment Effect on the Treated (ATT) on `current_cgpa` via 1:1 nearest-neighbour matching on logit-propensity (caliper = 0.2SD), with a 95% bootstrap CI (1,000 paired resamples). Implementation: `revision/code/causal_psm.py`.
3. **Honest interpretation.** The PSM result presented does *not* support a causal claim, and we now state this explicitly:

Quantitative result (Table 10).

Treatment	n treated	n matched	ATT	95 % CI	CI excl. 0?
Workshop participation	1,057	0	–	–	no acceptable matches
High peer-group quality ($\geq 4/5$)	117	10	+0.166	[−0.958, +1.417]	no
Stress coping (Yes vs No)	505	414	+0.081	[−0.062, +0.225]	no

Threats-to-Validity addendum (Section 5.9). “*SHAP attributions are associational and not, on their own, sufficient to support intervention prioritisation. The propensity-score-matched analysis in Section 5.5 provides a plausibility check showing that the directional signs of the SHAP-derived recommendations are consistent with directional ATT estimates after controlling for measured confounders, but observational matching cannot rule out unobserved confounding. Recommendations should therefore be interpreted as **hypothesis-generating decision support for academic advisors**, not as established causal interventions.*”

Future-work commitment (Section 6). A revised future-work paragraph commits to (1) cluster-randomised intervention trials at the institute level, (2) longitudinal multi-semester panels for DAG-based causal inference, (3) external validation across institutions and disciplines, (4) doubly-robust ATE estimators, and (5) interactive advisor dashboards with embedded fairness monitors.

[R2.3] Reproducibility insufficient

“Reproducibility is insufficient for a middle-tier empirical ML paper. Although the paper reports CV, SMOTE, Optuna, and some model settings, it omits the complete hyperparameter search spaces, final best parameters per objective, preprocessing mappings, feature-generation code, SMOTE parameters, fold indices, software versions, and exact availability conditions for the dataset.”

Reproduction in three commands. The reproducible implementation and experimental code are available at: <https://github.com/shikhapachouly/student-recommendation>

```
git clone https://github.com/shikhapachouly/student-recommendation.git
pip install -r requirements.txt
python -m src.pipeline --stage all          # produces all paper tables/figures
```

Software versions used in this revision Python 3.11.9, scikit-learn 1.5.2, XGBoost 2.1.4, PyTorch 2.4.1, pytorch-tabnet 4.1.0, SHAP 0.46.0, Optuna 4.8.0, imbalanced-learn 0.12.4, LightGBM 4.6.0, NumPy 1.26.4, pandas 2.2.3, matplotlib 3.9.4. SMOTE: `k_neighbors=5`, `random_state=42`, applied within each training fold only. Outer CV: `StratifiedKfold(n_splits=5, shuffle=True, random_state=42)`.

[R2.4] Target-leakage concern with academic_index

“CGPA and backlog-related variables are used to define academic and at-risk labels, while academic_index includes CGPA and prior academic marks. If CGPA or strongly label-defining variables remain in predictors for CGPA category or at-risk prediction, the task may become partially circular.”

Action. This is a legitimate and load-bearing concern. The leakage discipline was already enforced in code; we have now made it both more explicit and empirically validated.

Disciplines applied.

1. **current_cgpa** is excluded from **academic_index**. Only **tenth_percentage** and **twelfth_percentage** (prior academic record) are aggregated. Verifiable in `src/data/feature_engineering.py:36-50` and the inline comment in `src/config.py`.
2. **Target-defining columns are dropped** from the feature matrix for every objective: **current_cgpa**, **backlog_status**, **internship_completion**, **academic_projects_completed_yn**, **internship_payment_status**, **placement_status**, **higher_education_interest_yn**. Enforced in `src/data/preprocessor.py::encode_features`.
3. **engagement_score** excludes **internship_completion** and **academic_projects_completed_yn**, which define the career-readiness target.
4. **New Section 3.10 “Leakage Controls”** lists every excluded raw column per objective.
5. **New Table 3 “Leakage-free ablation”** reports F1-macro for each objective with and without prior academic features. Implementation: `revision/code/leakage_ablation.py`.

Quantitative result (Table 3).

Objective	F1-macro (full)	F1-macro (no prior academic)	Δ
CGPA Category	0.451 ± 0.027	0.387 ± 0.019	+0.063
At-Risk	0.639 ± 0.020	0.540 ± 0.034	+0.099
Career Ready	0.695 ± 0.021	0.588 ± 0.017	+0.107

Interpretation. Removing the three prior-academic predictors (**tenth_percentage**, **twelfth_percentage**, **academic_index**) drops F1-macro by 0.06–0.11 across the three objectives. The decline is the *legitimate* contribution of historical academic record — not circular leakage from the label. If the engineered **academic_index** were a residual echo of **current_cgpa**, removing it would collapse performance dramatically (a near-chance classifier); the moderate, non-collapsing decline confirms that the engineered features carry genuine prior-academic signal that is informative but *distinct* from the target. Source data: `revision/output/tables/leakage_ablation.csv`.

[R2.5] Fairness and subgroup analysis missing

“Gender appears among top SHAP features, but the paper does not evaluate subgroup performance, calibration, disparate error rates, or whether recommendations differ unfairly across demographic or socioeconomic groups. Since this is an educational decision-support system, fairness analysis is not optional.”

Action. A new **Section 5.7 “Fairness and Subgroup Analysis”** reports per-subgroup F1-macro, False-Negative Rate (FNR) for the positive class, Brier score, and the equalised-odds gap across **gender**, **residence_type**, and **family_income_bracket**. The audit re-uses the canonical 5-fold out of fold predictions, so the metrics are directly comparable to Table 12. A calibration plot for the at-risk objective by gender (new **Fig. 13**) is in `revision/output/figures/fairness_calibration.png`. Implementation: `revision/code/fairness.py`.

Quantitative result (Table 12 excerpt). The most consequential disparities are in the **family_income_bracket** attribute for the at-risk objective:

Obj.	Subgroup (income)	n	F1-macro	FNR	Brier
At-Risk	> 15 Lakh	60	0.704	0.429	0.159
At-Risk	8–15 Lakh	167	0.671	0.587	0.186
At-Risk	2.5–8 Lakh	414	0.674	0.548	0.192
At-Risk	< 2.5 Lakh	737	0.604	0.678	0.216
Equalised-odds gap ($\text{FNR}_{\max} - \text{FNR}_{\min}$) = 0.249					

Headline findings.

- **Income disparity for at-risk identification** is large: students from the lowest income bracket have an FNR of 0.678, versus 0.429 for the highest bracket — a gap of 0.249. The model misses low-income at-risk students more often, which is operationally the costliest error.
- **Gender gap for at-risk identification** is also substantive: female FNR = 0.715 vs. male FNR = 0.572 (gap = 0.143). The model is more likely to miss female at-risk students.
- **Calibration** is reasonable across subgroups (Brier scores 0.16–0.22), but the FNR gap dominates the operational risk.
- **Career-readiness** fairness is comparatively flat across all three attributes ($\Delta \text{FNR} \leq 0.13$).

Mitigation discussion (Section 5.7). We now state explicitly that (i) the appearance of **gender** among top SHAP features reflects regional dataset correlations, not causal influence; (ii) institutional deployments must monitor disparate impact via the equalised-odds gap and apply post-hoc reweighting when this gap exceeds an operationally chosen threshold; and (iii) recommendations are restricted to features marked *modifiable* in `MODIFIABLE_FEATURES`, never to immutable demographic features. The fairness numbers are tracked in `revision/output/tables/fairness_subgroups.csv`.

[R2.6] DL training fairness (insufficient detail)

“The comparison with deep learning models may not be fully fair. The dataset is small, but the manuscript does not provide enough detail on optimization, early stopping, batch size, epochs, learning-rate schedules, categorical embeddings, or regularization for TabNet, FT-Transformer, and SAINT.”

Action. A new **Section 4.3 “Deep-Learning Training Configuration”** lists every architectural and optimisation hyperparameter and confirms that the Optuna budget for each DL architecture is *identical* to that for XGBoost (50 trials $\times$ 3 inner folds = 150 inner-CV evaluations).

Setting	TabNet	FT-Transformer	SAINT
Optimiser	Adam	AdamW	AdamW
Initial learning rate	2×10^{-2}	1×10^{-3}	1×10^{-3}
LR scheduler	StepLR ($\gamma = 0.9$, step= 50)	Cosine	Cosine
Batch size	128	128	128
Maximum epochs	300	200	200
Early-stopping patience	20	20	20
In-fold validation split	15%	15%	15%
Categorical encoding	sparsemax mask	learned token	learned token
Architecture	$n_d = n_a = 8$, steps= 3, $\lambda_{\text{sparse}} = 10^{-3}$, $\gamma = 1.3$, momentum= 0.02, clip= 1.0	$d_{\text{token}} = 64$, $L = 2$, attention dropout= 0.2, FFN dropout= 0.1	$d = 32$, depth= 3, heads= 4, dropout= 0.1/0.1
Optuna trials (inner 3-fold)	50	50	50
Random seeds	42	42	42

Evidence (Table 6.)

3 Response to the Editor's Comments

Format compliance

Editor's item	Action
Body 11 pt	Applied.
Tables 10 pt	Applied.
Equations 11 pt italic	Applied; mathematical expressions are in italic font; * replaced by $\cdot$ or $\times$.
Wide figures one-column	Applied to Figs. 1, 3, 5–7, 9, 10, 13.
No two figures side-by-side	Verified.
No (a)/(b)/(c) labels in figures	Removed; moved to captions.
No borders around figures	Verified.
No rotated sheets / vertical writing	Verified.
Equations indented 5 mm with one blank line above/below	Applied.
Equation numbers right-aligned	Applied.
[1], [2–4] citation style with first-name initials	Reference list reformatted to IJIES sample style with <i>Vol.</i> , <i>No.</i> , <i>pp.</i> , year, doi.
All revisions in red font	Applied. The revised manuscript contains 331 red-font runs across 386 paragraphs (33.6 % of the text).
Page count ≥ 8	With the new Section 3.10, Section 4.3, Section 5.4, Section 5.5, Section 5.6, and Appendix A, the manuscript comfortably exceeds the 8-page minimum.

Closing Remarks

We have responded to every comment with concrete textual revision, supplementary experimental evidence, and reproducible code. Where the reviewer's substantive concern stands even after additional analysis (most notably for **[R2.2]**, the causal interpretation of SHAP-based recommendations), we have explicitly conceded the limitation and restated the manuscript's contribution accordingly, rather than overstating what the data supports. We believe this honest and transparent treatment best serves the readership and the methodological standards the reviewers raised.

We are deeply grateful for the editor's and the reviewers' time and care; their feedback materially improved this work.

Sincerely,

Shikha Pachouly (corresponding author)

Department of Computer Engineering, AISSMS College of Engineering, SPPU, Pune 411001, India

shikhaphd2022@gmail.com

D. S. Bormane

Department of Electronics and Telecommunication Engineering, AISSMS College of Engineering,
SPPU, Pune 411001, India